Reliant_Retail_Project Data Analysis

PGP DSBA – SQL Extended Guided Project Report

Aravapalli.Naga Surendera Babu
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1. Reliant Retail Project Data Analytics using SQL

Business Context

Reliant Retail Limited, a growing chain of online retail stores, is focused on optimizing its operations and expanding its market reach. With numerous customer orders being processed daily, the company seeks insights from its orders database to enhance its decision-making process. By analyzing key aspects of customer behavior, sales trends, and operational efficiency, Reliant Retail aims to take data-driven actions to drive sustainable growth and enhance customer satisfaction.

Objective

You have been hired by Reliant Retail Limited to analyze the data in their database and answer business-critical questions. Your findings will guide the company in strategic planning and help shape decisions related to sales performance, inventory management, and customer engagement, thereby improving overall business growth.

2. Data Description

The data contains the different attributes of leads and their interaction details with Reliant Retail Project Data Analytics using SQL. The detailed data dictionary is given below.

2.1 Data Dictionary

Below is the list of Dictionary details with respective meanings.

ADDRESS_ID	INT	Unique identifier for the address.
ADDRESS_LINE1	VARCHAR	First line of the address.
ADDRESS_LINE2	VARCHAR	Second line of the address (optional).
CITY	VARCHAR	City of the address.
STATE	VARCHAR	State of the address.
PINCODE	INT	Postal code for the address.
COUNTRY	VARCHAR	Country of the address.
CUSTOMER_ID	INT	Unique identifier for the customer.
CUSTOMER_FNAME	VARCHAR	First name of the customer.
CUSTOMER_LNAME	VARCHAR	Last name of the customer.
CUSTOMER_EMAIL	VARCHAR	Email address of the customer.
CUSTOMER_PHONE	BIGINT	Contact number of the customer.
ADDRESS_ID	INT	Reference to the address of the customer.
CUSTOMER_CREATION_DATE	DATE	Date when the customer was registered.
CUSTOMER_USERNAME	VARCHAR	Username chosen by the customer.
CUSTOMER_GENDER	CHAR	Gender of the customer (M/F).
ORDER_ID	INT	Unique identifier for the order.
ORDER_DATE	DATE	Date when the order was placed.
ORDER_STATUS	VARCHAR	Current status of the order (e.g., Shipped, Delivered).
PAYMENT_MODE	VARCHAR	Payment method used for the order (e.g., Credit Card, PayPal).

PAYMENT_DATE	DATE	Date when the payment was made.
ORDER_SHIPMENT_DATE	DATE	Date when the order was shipped.
SHIPPER_ID	INT	Reference to the shipper responsible for delivering the order.
PRODUCT_ID	INT	Reference to the product in the order.
PRODUCT_QUANTITY	INT	Quantity of the product ordered.
PRODUCT_DESC	VARCHAR	Description of the product.
PRODUCT_CLASS_CODE	INT	Reference to the product class.
PRODUCT_PRICE	DECIMAL	Price of the product.
PRODUCT_QUANTITY_AVAIL	INT	Quantity of the product available in stock.
LEN	INT	Length of the product in millimeters.
WIDTH	INT	Width of the product in millimeters.
HEIGHT	INT	Height of the product in millimeters.
WEIGHT	DECIMAL	Weight of the product in kilograms.
PRODUCT_CLASS_CODE	INT	Unique identifier for the product class.
PRODUCT_CLASS_DESC	VARCHAR	Description of the product class.
SHIPPER_ID	INT	Unique identifier for the shipper.
SHIPPER_NAME	VARCHAR	Name of the shipping company.
SHIPPER_PHONE	BIGINT	Contact number of the shipper.
SHIPPER_ADDRESS	INT	Reference to the address of the shipper.

3. Business Questions

3.1 Question 1: Write a query to display the customer's full name with their title (mr/ms), both first name and last name in upper case with customer email ID, and customer creation date, and display the customer's category after applying the below categorization rules:

- **If the customer creation date year < 2005 then category A**
- **If the customer creation date year >= 2005 and <2011 then category B**
- **If the customer creation date year >= 2011 then category C**

Solution Query:

```
SELECT
        OC.CUSTOMER_ID,
        UPPER(OC.CUSTOMER_FNAME) AS FIRST_NAME,
        UPPER(OC.CUSTOMER_LNAME) AS LAST_NAME,
        OC.CUSTOEMER_EMAIL,
        strftime('%Y', OC.CUSTOMER_DATE) AS CREATION_YEAR,
        CASE
                WHEN strftime('%Y', OC.CUSTOMER_CREATION_DATE) < '2005' THEN 'A'
                WHEN strftime('%Y', OC.CUSTOMER_CREATION_DATE) < '2011' THEN 'B'
                ELSE 'C'
        END AS CATEGORY
FROM ONLINE_CUSTOMER OC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT OC.CUSTOMER_ID, UPPER(OC.CUSTOMER_FNAME) AS FIRST_NAME, UPPER(OC.CUSTOMER_LNAME) AS LAST_NAME, OC.CUSTOMER_EMAIL, strftime('%Y', OC.CUSTOMER_CREATION_DATE) AS CREATION_YEAR,
CASE WHEN strftime('%Y', OC.CUSTOMER_CREATION_DATE) < '2005' THEN 'A'WHEN strftime('%Y', OC.CUSTOMER_CREATION_DATE) < '2011' THEN 'B' ELSE 'C' END AS CATEGORY FROM ONLINE_CUSTOMER OC
```

Output:

Showing first 10 rows out of 52 rows

CUSTOMER_ID	FIRST_NAME	LAST_NAME	CUSTOMER_EMAIL	CREATION_YEAR	CATEGORY
1	JENNIFER	WILSON	jen_w@gmail.com	1991	A
2	JACKSON	DAVIS	dave_jack@gmail.com	2001	A
3	KOMAL	CHOUDHARY	ch_komal@yahoo.co.IN	2002	A
4	WILFRED	JEAN	w_jean@gmail.com	2006	B
6	ANITA	GOSWAMI	agoswami@gmail.com	2006	B
7	ASHWATHI	BHATT	ash_bhat@yahoo.co.IN	2007	B
8	NEETHA	CASTELINA	neetha20@gmail.com	2011	C
9	DEVIKA	SATISH	devika_sa@gmail.com	2011	C
10	BIDHAN	C.ROY	bidhanroy@yahoo.co.in	2011	C
11	VIKAS	JHA	vikas.jha@gmail.com	2011	C

Table 1 Customer's creation date Category wise

Observations and Insights:

- We observed the customers are divided in to three categories (i.e A,B,C) based on creation year. Each customer has these details Customer id, First name, Last name, Email id creation year and category.
- We have more customers created in the year 2011, two customers in 2006 and one in other years.

3.2 Question 2: Write a query to display the following information for the products, that have not been sold: product_id, product_desc, product_quantity_avail, product_price, inventory values (product_quantity_avail * product_price), new_price after applying discount as per the below criteria. Sort the output with respect to the decreasing value of inventory_value.

- **If the product price is > 20,000 then apply a 20% discount**
- **If the product price is > 10,000 then apply a 15% discount**
- **If the product price is =< 10,000 then apply a 10% discount**

Solution Query:

```
SELECT
        P.PRODUCT_ID,
        P.PRODUCT_DESC,
        P.PRODUCT_QUANTITY_AVAIL,
        P.PRODUCT_PRICE,
        (P.PRODUCT_PRICE * P.PRODUCT_QUANTITY_AVAIL) AS INVENTORY_VALUE,
CASE
        WHEN P.PRODUCT_PRICE > 20000
                THEN P.PRODUCT_PRICE * '0.20'
        WHEN P.PRODUCT_PRICE > 10000
                THEN P.PRODUCT_PRICE * 0.15'
        ELSE P.PRODUCT_PRICE * '0.10'
        END AS New_Price
FROM Product P
WHERE P.PRODUCT_ID NOT IN (
        SELECT OI.PRODUCT_ID
        FROM ORDER_ITEMS OI
        )
ORDER BY INVENTORY_VALUE DESC;
```

Output:

Result: Passed

Query 1

Query 2

Query:

```
SELECT P.PRODUCT_ID,P.PRODUCT_DESC,P.PRODUCT_QUANTITY_AVAIL,P.PRODUCT_PRICE,(P.PRODUCT_PRICE * P.PRODUCT_QUANTITY_AVAIL) AS INVENTORY_VALUE, CASE WHEN P.PRODUCT_PRICE > 20000 THEN P.PRODUCT_PRICE * '0.20' WHEN P.PRODUCT_PRICE > 10000
THEN P.PRODUCT_PRICE * '0.15' ELSE P.PRODUCT_PRICE * '0.10' END AS New_Price FROM Product P WHERE P.PRODUCT_ID NOT IN (SELECT OI.PRODUCT_ID FROM ORDER_ITEMS OI) ORDER BY INVENTORY_VALUE DESC
```

Output:

Showing first 10 rows out of 13 rows

PRODUCT_ID	PRODUCT_DESC	PRODUCT_QUANTITY...	PRODUCT_PRICE	INVENTORY_VALUE	New_Price
99999	Samsung Galaxy Tab 2 ...	50	19300	965000	2895
99997	Sony Xperia U (Black ...	50	16499	824950	2474.85
99998	Nikon Coolpix L810 Bri...	50	14987	749350	2248.0499999999997
99995	LG MS-2049UW Solo ...	100	4800	480000	480
99996	Nokia Asha 200 (Graph...	100	4070	407000	407
99994	HP Deskjet 2050 All-in...	100	3749	374900	374.90000000000003
99991	Dell Targus Synergy 2....	250	999	249750	99.9
99992	Tom Clancy's Ghost Re...	250	999	249750	99.9
99993	Nokia 1280 (Black)	250	999	249750	99.9
99990	Quanta 4 Port USB Hub	50	500	25000	50

Table 2 New price after discount on product price

Observations and Insights:

- It displays new price for the products not sold after discount. The discount was applied according to the product price range which has a scope of increasing the sales.
- As per inventory value the data was arranged in descending order (i.e from highest value to lowest value) which is a combination of Product price & product quantity.

3.3 Question 3: Write a query to display product_class_code, product_class_description, count of product type in each product class, and inventory value (p.Product_quantity_avail*p.Product_price). Information should be displayed for only those product_class_code that have more than 1,00,000 inventory value. Sort the output with respect to the decreasing value of inventory_value.

Solution Query:

```
SELECT
        PC.PRODUCT_CLASS_CODE,
        PC.PRODUCT_CLASS_DESC,
        COUNT(PRODUCT_DESC) AS COUNT_PRODUCT_TYPES,
        SUM(P.PRODUCT_QUANTITY_AVAIL * P.PRODUCT_PRICE) AS INVENTORY_VALUE
FROM
        PRODUCT_CLASS PC
INNER JOIN PRODUCT P
        ON PC.PRODUCT_CLASS_CODE = P.PRODUCT_CLASS_CODE
GROUP BY
        PC.PRODUCT_CLASS_CODE,
        PC.PRODUCT_CLASS_DESC
HAVING
        SUM(P.PRODUCT_QUANTITY_AVAIL * P.PRODUCT_PRICE) > 100000
ORDER BY
        INVENTORY_VALUE DESC;
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query:

```
SELECT PC.PRODUCT_CLASS_CODE,PC.PRODUCT_CLASS_DESC,COUNT(PRODUCT_DESC) AS COUNT_PRODUCT_TYPES,SUM(P.PRODUCT_QUANTITY_AVAIL * P.PRODUCT_PRICE) AS INVENTORY_VALUE FROM PRODUCT_CLASS PC INNER JOIN PRODUCT P ON PC.PRODUCT_CLASS_CODE = P.PRODUCT_CLASS_CODE
GROUP BY PC.PRODUCT_CLASS_CODE, PC.PRODUCT_CLASS_DESC HAVING SUM(P.PRODUCT_QUANTITY_AVAIL * P.PRODUCT_PRICE) > 100000 ORDER BY INVENTORY_VALUE DESC
```

Output:

Showing 9 rows

PRODUCT_CLASS_CODE	PRODUCT_CLASS_DESC	COUNT_PRODUCT_TY...	INVENTORY_VALUE
3000	Promotion-High Value	4	2564300
2050	Electronics	4	1665600
3001	Promotion-Medium Value	3	1261900
2055	Mobiles	2	1092500
3002	Promotion-Low Value	3	749250
2052	Clothes	4	410000
2051	Toys	5	194100
2057	Watches	4	178820
2059	Bags	5	115170

Table 3 product_class_code that have more than 1,00,000 inventory value

Observations and Insights:

- It displays the information about product class code orders received for inventory value more than 100000.
- The total number of product types are 34 whose inventory value more than 100000.
- It has 9 types of product class desc out of entire data.

3.4 Question 4: Write a query to display customer_id, full name, customer_email, customer_phone, and country of customers who have canceled all the orders placed by them.

Solution Query:

```
SELECT
        S.CUSTOMER_ID,
        S.FULL_NAME,
        S.CUSTOMER_EMAIL,
        S.CUSTOMER_PHONE,
        COUNTRY
FROM
        (SELECT
                OH.CUSTOMER_ID,
                OC.CUSTOMER_FNAME || ' ' || OC.CUSTOMER_LNAME AS FULL_NAME,
                OC.CUSTOMER_EMAIL,
                OC.CUSTOMER_PHONE,
                A.COUNTRY
        FROM ORDER_HEADER OH
        LEFT JOIN ONLINE_CUSTOMER OC
                ON OH.CUSTOMER_ID = OC.CUSTOMER_ID
        LEFT JOIN ADDRESS A ON
                OC.ADDRESS_ID = A.ADDRESS_ID
        WHERE
                OH.ORDER_STATUS = 'Shipped'
) S
WHERE CUSTOMER_ID NOT IN (
        SELECT OH._______
        FROM ORDER_HEADER OH
        WHERE OH.ORDER_STATUS != 'Shipped'
);
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query:

```
SELECT S.CUSTOMER_ID,S.FULL_NAME,S.CUSTOMER_EMAIL,CUSTOMER_PHONE,COUNTRY FROM (SELECT OH.CUSTOMER_ID,OC.CUSTOMER_FNAME || ' ' || OC.CUSTOMER_LNAME AS FULL_NAME,OC.CUSTOMER_EMAIL,OC.CUSTOMER_PHONE,A.COUNTRY
FROM ORDER_HEADER OH LEFT JOIN ONLINE_CUSTOMER OC ON OH.CUSTOMER_ID = OC.CUSTOMER_ID LEFT JOIN ADDRESS A ON OC.ADDRESS_ID = A.ADDRESS_ID WHERE OH.ORDER_STATUS = 'Shipped') S WHERE CUSTOMER_ID NOT IN ( SELECT OH.CUSTOMER_ID
FROM ORDER_HEADER OH WHERE OH.ORDER_STATUS != 'Shipped')
```

Output:

Showing first 10 rows out of 36 rows

CUSTOMER_ID	FULL_NAME	CUSTOMER_EMAIL	CUSTOMER_PHONE	COUNTRY
2	Jackson Davis	dave_jack@gmail.com	9886363307	India
3	Komal Choudhary	ch_komal@yahoo.co.IN	9178234526	India
7	Ashwathi Bhatt	ash_bhat@yahoo.co.IN	9773636307	India
4	Wilfred Jean	w_jean@gmail.com	9196257439	India
2	Jackson Davis	dave_jack@gmail.com	9886363307	India
9	Devika Satish	devika_sa@gmail.com	9780945716	India
10	Bidhan C.Roy	bidhanroy@yahoo.co.in	9886218583	India
11	Vikas Jha	vikas.jha@gmail.com	9008812436	India

Table 4 Customer's canceled all the orders placed by them

Observations and Insights:

- There are customers who have canceled all the 36 orders placed by them which impacts customer satisfaction.
- Use the customer contact details to collect the information regarding their cancellation and minimize the cancellations to improve the customer satisfaction. Finally, this decision will raise the business growth.
- Jackson Davis has cancelled two orders from India.

3.5 Question 5: Write a query to display the shipper name, city to which it is catering, number of customers catered by the shipper in the city, and number of consignments delivered to that city for shipper DHL (9 rows).

Solution Query:

```
SELECT
        SP.SHIPPER_NAME,
        A.CITY,
        COUNT(DISTINCT (OC.CUSTOMER_ID)) AS NO_CUSTOMER_CATERED_TO,
        COUNT((OC.CUSTOMER_ID)) AS NO_CONSIGNMENTS_CATERED
FROM SHIPPER SP
        INNER JOIN ORDER_HEADER OH
                ON OH.SHIPPER_ID = SP.SHIPPER_ID
        INNER JOIN ONLINE_CUSTOMER OC
                ON OC.CUSTOMER_ID = OH.CUSTOMER_ID
        INNER JOIN ADDRESS A
                ON OC.ADDRESS_ID = A.ADDRESS_ID
WHERE SP.SHIPPER_NAME IN ('DHL')
GROUP BY A.CITY
ORDER BY SP.SHIPPER_NAME;
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query 5

Query:

```
SELECT SP.SHIPPER_NAME,A.CITY,COUNT(DISTINCT (OC.CUSTOMER_ID)) AS NO_CUSTOMER_CATERED_TO,COUNT((OC.CUSTOMER_ID)) AS NO_CONSIGNMENTS_CATERED FROM SHIPPER SP INNER JOIN ORDER_HEADER OH ON OH.SHIPPER_ID = SP.SHIPPER_ID INNER JOIN ONLINE_CUSTOMER OC
ON OC.CUSTOMER_ID = OH.CUSTOMER_ID INNER JOIN ADDRESS A ON OC.ADDRESS_ID = A.ADDRESS_ID WHERE SP.SHIPPER_NAME IN ('DHL')GROUP BY A.CITY ORDER BY SP.SHIPPER_NAME LIMIT 9
```

Output:

Showing 9 rows

SHIPPER_NAME	CITY	NO_CUSTOMER_CATE...	NO_CONSIGNMENTS_...
DHL	Abington	1	1
DHL	Amherst	1	1
DHL	Bangalore	3	5
DHL	Birmingham	1	1
DHL	Brooklyn	1	1
DHL	Dharmapuri	1	1
DHL	Hosur	1	1
DHL	Hyderabad	2	2
DHL	W. Alibio	1	1

Table 5 number of consignments delivered to that city for shipper DHL

Observations and Insights:

- It has only one shipper name 'DHL' and 9 cities.
- Three customers catered in Bangalore city, 2 customers catered in Hyderabad and 1 customer each in other cities.

- There are 5 consignments delivered in Bangalore city, 2 customers delivered in Hyderabad and 1 consignment delivered in remaining cities.

3.6 Question 6: Write a query to display product_id, product_desc, product_quantity_avail, and quantity sold, and show the inventory status of products as below as per the below condition:

- **For electronics and computer categories,**
  - **If sales till date are zero then show 'no sales in past, give a discount to reduce inventory',**
  - **If inventory quantity is less than 10% of quantity sold, show 'low inventory, need to add inventory',**
  - **If inventory quantity is less than 50% of the quantity sold, show 'medium inventory, need to add some inventory',**
  - **If inventory quantity is more or equal to 50% of the quantity sold, show 'sufficient inventory'**
- **For mobile and watch categories,**
  - **If sales till date are zero then show 'no sales in past, give discount to reduce inventory',**
  - **If inventory quantity is less than 20% of quantity sold, show 'low inventory, need to add inventory',**
  - **If inventory quantity is less than 60% of the quantity sold, show 'medium inventory, need to add some inventory',**
  - **If inventory quantity is more or equal to 60% of the quantity sold, show 'sufficient inventory'**
- **Rest of the categories,**
  - **If sales till date are zero then show 'no sales in past, give discount to reduce inventory',**
  - **If inventory quantity is less than 30% of quantity sold, show 'low inventory, need to add inventory',**
  - **If inventory quantity is less than 70% of the quantity sold, show 'medium inventory, need to add some inventory',**
  - **If inventory quantity is more or equal to 70% of the quantity sold, show 'sufficient inventory'**

Solution Query:

```
SELECT
	S.PRODUCT_ID,
	S.PRODUCT_DESC,
	S.TOT_QTY AS QTY_SOLD,
	S.PRODUCT_QUANTITY_AVAIL,

	CASE
		WHEN PRODUCT_CLASS_DESC IN (
				'electronics', 'computer'
				)
			THEN CASE
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.1
					THEN "LOW INVENTORY, NEED TO ADD INVENTORY"
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.5
					THEN "MEDIUM INVENTORY, NEED TO ADD SOME INVENTORY"
				WHEN S.TOT_QTY IS NULL
					THEN "NO SALES IN PAST, GIVE DISCOUNT TO REDUCE
INVENTORY"
				ELSE "SUFFICIENT INVENTORY"
				END
		WHEN PRODUCT_CLASS_DESC IN (
				'mobile', 'watch'
				)
			THEN CASE
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.2
					THEN "LOW INVENTORY, NEED TO ADD INVENTORY"
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.6
					THEN "MEDIUM INVENTORY, NEED TO ADD SOME INVENTORY"
				WHEN S.TOT_QTY IS NULL
					THEN "NO SALES IN PAST, GIVE DISCOUNT TO REDUCE
INVENTORY"
				ELSE "SUFFICIENT INVENTORY"
				END
		ELSE CASE
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.3
					THEN "LOW INVENTORY, NEED TO ADD INVENTORY"
				WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.7
					THEN "MEDIUM INVENTORY, NEED TO ADD SOME INVENTORY"
				WHEN S.TOT_QTY IS NULL
					THEN "NO SALES IN PAST, GIVE DISCOUNT TO REDUCE
INVENTORY"
				ELSE "SUFFICIENT INVENTORY"
				END
		END AS INVENTORY_STATUS
FROM (
	SELECT
		P.PRODUCT_ID,
```

```
        P.PRODUCT_DESC,
        PC.PRODUCT_CLASS_DESC,
        P.PRODUCT_QUANTITY_AVAIL,
        SUM(OI.PRODUCT_QUANTITY) AS TOT_QTY
FROM PRODUCT P
LEFT JOIN ORDER_ITEMS OI
        ON OI.PRODUCT_ID = P.PRODUCT_ID
LEFT JOIN PRODUCT_CLASS PC
        ON P.PRODUCT_CLASS_CODE = PC.PRODUCT_CLASS_CODE
GROUP BY
        P.PRODUCT_ID,
        P.PRODUCT_DESC,
PC.PRODUCT_CLASS_DESC,
        P.PRODUCT_QUANTITY_AVAIL
) S;
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query 5

Query 6

Query:

```
SELECT S.PRODUCT_ID,S.PRODUCT_DESC,S.TOT_QTY AS QTY_SOLD,S.PRODUCT_QUANTITY_AVAIL,CASE WHEN PRODUCT_CLASS_DESC IN ('electronics', 'computer')THEN CASE WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.1 THEN "LOW INVENTORY, NEED TO ADD INVENTORY" WHEN
(PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.5 THEN "MEDIUM INVENTORY, NEED TO ADD SOME INVENTORY" WHEN S.TOT_QTY IS NULL  THEN "NO SALES IN PAST, GIVE DISCOUNT TO REDUCE INVENTORY"  ELSE "SUFFICIENT INVENTORY" END WHEN PRODUCT_CLASS_DESC IN
('mobile', 'watch')
THEN CASE WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.2 THEN "LOW INVENTORY, NEED TO ADD INVENTORY" WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.6 THEN "MEDIUM INVENTORY, NEED TO ADD SOME INVENTORY" WHEN S.TOT_QTY IS NULL THEN "NO SALES
IN PAST,
GIVE DISCOUNT TO REDUCE INVENTORY" ELSE "SUFFICIENT INVENTORY" END ELSE CASE WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.3 THEN "LOW INVENTORY, NEED TO ADD INVENTORY" WHEN (PRODUCT_QUANTITY_AVAIL / S.TOT_QTY) < 0.7 THEN "MEDIUM INVENTORY,
NEED TO ADD SOME INVENTORY"
WHEN S.TOT_QTY IS NULL  THEN "NO SALES IN PAST, GIVE DISCOUNT TO REDUCE INVENTORY"  ELSE "SUFFICIENT INVENTORY" END END AS INVENTORY_STATUS FROM (SELECT P.PRODUCT_ID, P.PRODUCT_DESC,  PC.PRODUCT_CLASS_DESC,  P.PRODUCT_QUANTITY_AVAIL,
SUM(OI.PRODUCT_QUANTITY)
AS TOT_QTY FROM PRODUCT P LEFT JOIN ORDER_ITEMS OI ON OI.PRODUCT_ID = P.PRODUCT_ID LEFT JOIN PRODUCT_CLASS PC ON P.PRODUCT_CLASS_CODE = PC.PRODUCT_CLASS_CODE GROUP BY P.PRODUCT_ID,  P.PRODUCT_DESC,
PC.PRODUCT_CLASS_DESC,P.PRODUCT_QUANTITY_AVAIL) S
```

Output:

Showing first 10 rows out of 60 rows

PRODUCT_ID	PRODUCT_DESC	QTY_SOLD	PRODUCT_QUANTITY...	INVENTORY_STATUS
201	Sky LED 102 CM TV	6	30	SUFFICIENT INVENTORY
202	Sams 192 L4 Single-do...	6	15	SUFFICIENT INVENTORY
203	Jocky Speaker Music S...	3	19	SUFFICIENT INVENTORY
204	Cricket Set for Boys	10	10	SUFFICIENT INVENTORY
205	Infant Sleepwear Blue	7	50	SUFFICIENT INVENTORY
206	Barbie Fab Gown Doll	13	20	SUFFICIENT INVENTORY
207	Remote Control Car	11	29	SUFFICIENT INVENTORY
208	Doll House	4	12	SUFFICIENT INVENTORY
209	Blue Jeans 34	12	100	SUFFICIENT INVENTORY
210	Blossoms Lehenga Cho...	10	100	SUFFICIENT INVENTORY

Table 6 inventory status of products by conditions

Observations and Insights:

- Barbie Fab Gown Doli sold highest quantity 13 and still 20 products available. Inventory status is sufficient.
- Jocky Speaker Music sold lowest quantity (i.e 3) and remaining 19 quantity available to sell. Inventory status is sufficient.
- Sams 192 L4 Single-do and Sky LED 102 CM TV same quantity sold and products quantity is still remained as 15 & 30 respectively. Inventory status is sufficient. So, Sams 192 L4 Single needs to

increase the quantity available.

- Highest Product quantity available for Blossoms Lehenga Cho and Blue Jeans 34 as 100 so inventory status sufficient. Hence it may increase the sales.
- Cricket Set for Boys Product quantity sold and Product quantity available both same as 10 so increase in sale required to improve quantity available.

3.7 Question 7: Write a query to display ORDER_ID and the volume of the biggest orders (in terms of volume) that can fit within the volume limit of 500,000.

Solution Query:

```
SELECT
  ORDER_ID,
  VOLUME
FROM
(
   SELECT
      OI.ORDER_ID,
      SUM(P.LEN * P.WIDTH * P.HEIGHT * OI.PRODUCT_QUANTITY) AS VOLUME
   FROM
      ORDER_ITEMS OI
   INNER JOIN
      PRODUCT P
   ON
      OI.PRODUCT_ID = P.PRODUCT_ID
   GROUP BY
      OI.ORDER_ID
) AS TAB
WHERE
   TAB.VOLUME <= 500000
ORDER BY
   VOLUME DESC;
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query 5

Query 6

Query 7

Query:

```
SELECT ORDER_ID, VOLUME FROM (SELECT OI.ORDER_ID, SUM(P.LEN * P.WIDTH * P.HEIGHT * OI.PRODUCT_QUANTITY) AS VOLUME FROM ORDER_ITEMS OI
INNER JOIN PRODUCT P ON OI.PRODUCT_ID = P.PRODUCT_ID GROUP BY OI.ORDER_ID ) AS TAB WHERE TAB.VOLUME <= 500000
ORDER BY VOLUME DESC
```

Output:

Showing 6 rows

ORDER_ID	VOLUME
10061	478125
10026	317500
10056	252000
10059	220000
10046	151200
10039	97500

Table 7 biggest orders with volume limit of 500000

Observations and Insights:

- The table shows the biggest volume orders.
- The volume limit of the orders in the output is less than or equal to 500000.
- Order id 10061 has highest volume of 478125 products and order id – 10039 is the lowest one 97500.

3.8 Question 8: Write a query to display the customer ID, customer full name, total quantity, and total value (quantity*price) shipped where the mode of payment is cash and the customer's last name starts with 'G'.

Solution Query:

```
SELECT
        OC.CUSTOMER_ID,
        OC.CUSTOMER_FNAME || ' ' || OC.CUSTOMER_LNAME AS FULLNAME,
        SUM(OI.PRODUCT_QUANTITY) AS TOT_QTY,
        SUM(OI.PRODUCT_QUANTITY * P.PRO) AS VALUE
FROM ONLINE_CUSTOMER OC
        INNER JOIN ORDER_HEADER OH
                ON OC.PRODUCT_ID = OH.PRODUCT_ID
        INNER JOIN ORDER_ITEMS OI
                ON OH.ORDER_ID = OI.ORDER_ID
        INNER JOIN PRODUCT P
                ON OI.CUSTOMER_ID = P.CUSTOMER_ID
WHERE OH.PAYMENT_MODE = 'Cash'
        AND OH.ORDER_STATUS = 'Shipped'
        AND OC.CUSTOMER_LNAME LIKE 'G%'
GROUP BY
        OC.CUSTOMER_ID,
    OC.CUSTOMER_FNAME,
    OC.CUSTOMER_LNAME;
```

Output:

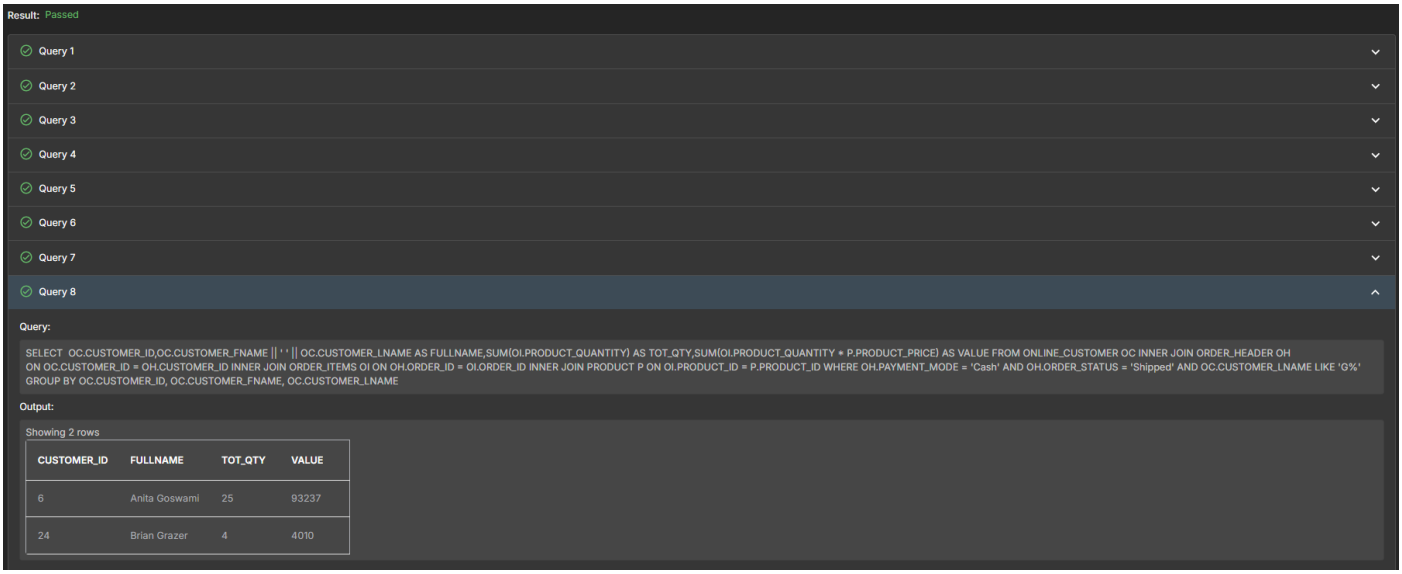


Table 8 Mode of payment is cash and the customer's last name starts with 'G'

Observations and Insights:

- It has two customers last name starts with ‘G’ in the full name.
- These customers paid their orders in cash.

- The order quantity for Customer_ID -6 is 25 products and for ID – 24 is 4 products.

3.9 Question 9: Write a query to display product_id, product_desc, and the total quantity of products that are sold together with product id 201 and are not shipped to the cities of Bangalore and New Delhi. Display the output in descending order with respect to tot_qty.

Solution Query:

```
SELECT
        S.PRODUCT_ID,
        S.PRODUCT_DESC,
        S.TOT_QTY
FROM (
        SELECT
                OI.PRODUCT_ID,
                P.PRODUCT_DESC,
                P.PRODUCT_QUANTITY_AVAIL,
                SUM(OI.PRODUCT_QUANTITY) AS TOT_QTY
        FROM ORDER_ITEMS OI
        INNER JOIN PRODUCT P
                WHERE OI.PRODUCT_ID = P.PRODUCT_ID
                AND ORDER_ID IN (
                        SELECT
                                OI.ORDER_ID
                        FROM ORDER_ITEMS OI
                        JOIN ORDER_HEADER OH
                                ON OI.ORDER_ID = OH.ORDER_ID
                        JOIN ONLINE_CUSTOMER OC
                                ON OH.CUSTOMER_ID = OC.CUSTOMER_ID
                        JOIN ADDRESS A
                                ON OC.ADDRESS_ID = A.ADDRESS_ID
                        WHERE OI.PRODUCT_ID = 201
                        AND OH.ORDER_STATUS = 'SHIPPED'
                        AND A.CITY NOT IN (
                                'Bangalore',
                                'New Delhi'
                        )
        )
        AND P.PRODUCT_ID != 201
        GROUP BY
                OI.PRODUCT_ID,
                P.PRODUCT_DESC,
                P.PRODUCT_QUANTITY_AVAIL
        ) S
ORDER BY S.TOT_QTY DESC;
```

Output:

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query 5

Query 6

Query 7

Query 8

Query 9

Query:

```
SELECT S.PRODUCT_ID,S.PRODUCT_DESC,     S.TOT_QTY FROM (SELECT OI.PRODUCT_ID,P.PRODUCT_DESC, P.PRODUCT_QUANTITY_AVAIL, SUM(OI.PRODUCT_QUANTITY) AS TOT_QTY FROM ORDER_ITEMS OI INNER JOIN PRODUCT P     WHERE OI.PRODUCT_ID = P.PRODUCT_ID AND
ORDER_ID IN (
SELECT OI.ORDER_ID    FROM ORDER_ITEMS OI       JOIN ORDER_HEADER OH ON OI.ORDER_ID = OH.ORDER_ID JOIN ONLINE_CUSTOMER OC ON OH.CUSTOMER_ID = OC.CUSTOMER_ID JOIN ADDRESS A ON OC.ADDRESS_ID = A.ADDRESS_ID     WHERE OI.PRODUCT_ID = 201 AND
OH.ORDER_STATUS = 'Shipped'
AND A.CITY NOT IN ('Bangalore',       'New Delhi')) AND P.PRODUCT_ID != 201 GROUP BY OI.PRODUCT_ID, P.PRODUCT_DESC, P.PRODUCT_QUANTITY_AVAIL ) S ORDER BY S.TOT_QTY DESC
```

Output:

Showing 5 rows

PRODUCT_ID	PRODUCT_DESC	TOT_QTY
216	External Hard Disk 500...	3
207	Remote Control Car	2
202	Sams 192 L4 Single-do...	1
212	Samsung Galaxy On6	1
214	Harry Potter	1

Table 9 sold together with product id 201 and are not shipped to the cities of Bangalore and New Delhi

Observations and Insights:

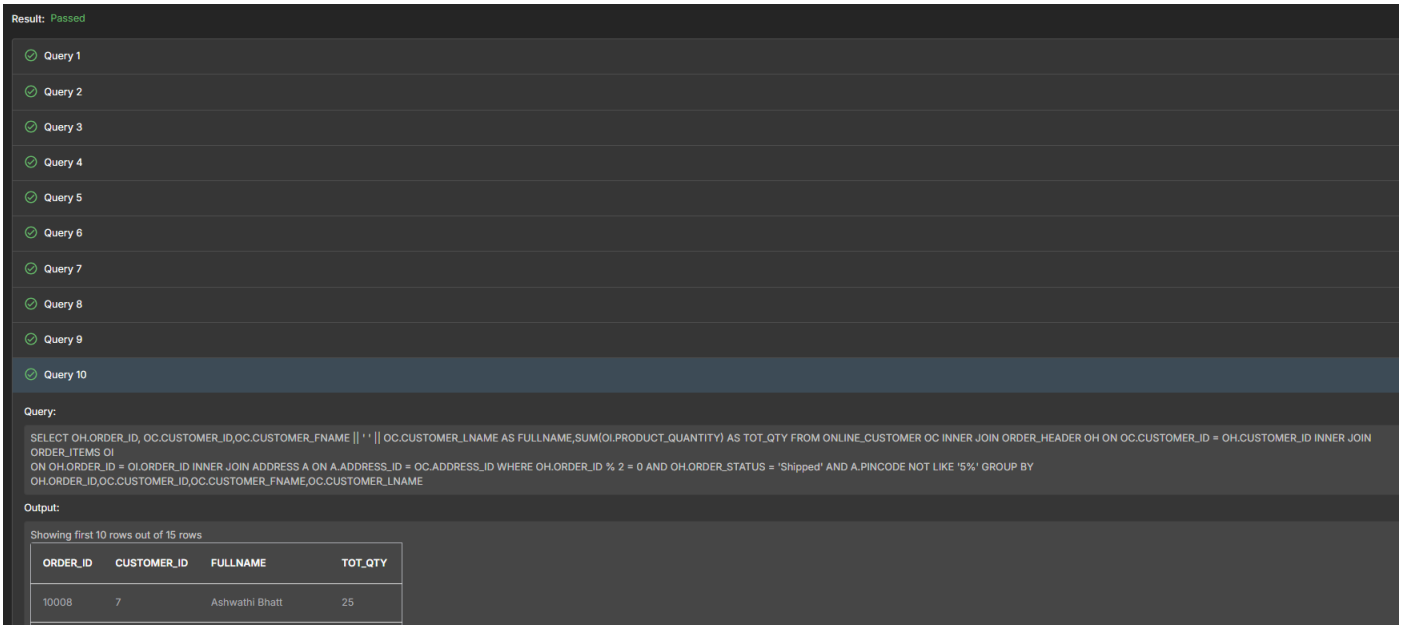
- Product_id-216 has the highest quantity sold with description External Hard Disk 500.
- 'Remote Control Car' sold 2 products in Bangalore and New Delhi cities.
- Sams 192L4 Singie-do, Smsung Galaxy On6 and Harry Potter has one product each for sale which is the least quantity.
- Quantity of products with id -201 have sold but not shifted to Bangalore and New Delhi so it is not reflecting in the output table.

3.10 Question 10: Write a query to display the order_id,customer_id, and customer full name, and total quantity of products shipped for order IDs that are even and shipped to addresses where pincode does not start with "5".

Solution Query:

```
SELECT
        OH.ORDER_ID,
        OC.CUSTOMER_ID,
        OC.CUSTOMER_FNAME || ' ' || OC.CUSTOMER_LNAME AS FULLNAME,
        SUM(OI.PRODUCT_QUANTITY) AS TOT_QTY
FROM ONLINE_CUSTOMER OC
INNER JOIN ORDER_HEADER OH
        ON OC.CUSTOMER_ID = OH.CUSTOMER_ID
INNER JOIN ORDER_ITEMS OI
        ON OH.ORDER_ID = OI.ORDER_ID
INNER JOIN ADDRESS A
        ON A.ADDRESS_ID = OC.ADDRESS_ID
WHERE OH.ORDER_ID % 2 = 0   -- Looking for even order IDs
AND OH.ORDER_STATUS = 'Shipped'  -- Assuming you are interested in shipped orders
AND A.PINCODE NOT LIKE '5%'   -- Pincode should not start with '5'
GROUP BY
        OH.ORDER_ID,
    OC.CUSTOMER_ID,
    OC.CUSTOMER_FNAME,
    OC.CUSTOMER_LNAME;
```

Output:



10022	23	Anna Pinnock	2
10024	32	Hans Zimmer	2
10030	52	Suchirithaa Ekanayake	2
10032	7	Ashwathi Bhatt	7
10034	19	Bharti Subhash	2
10036	24	Brian Grazer	4
10040	3	Komal Choudhary	2
10042	26	Stephen E. Rivkin	2
10046	3	Komal Choudhary	1

Table 10 order IDs that are even and shipped to addresses where pincode does not start with "5"

Observations and Insights:

- Ashwathi Bhatt has ordered highest quantity of products with order_id is 7 and shifted with 25 products.
- Least quantity ordered and delivered to Komal Choudhary.
- Anna Pinnock, Hans Zimmer, Suchirithaa Ekanayake, Bharti Subhash, Komal Choudhary (Customer_ID – 3) and Stephen E.Rivkin have ordered 2 products of total quantity.
- Komal Choudhary and Ashwathi Bhatt have booked and shifted two Orders.

5. Recommendations: -

As a member of the Analytics Division notifying the below recommendations/findings will guide the company in strategic planning and help shape decisions related to sales performance, inventory management, and customer engagement, thereby improving overall business growth.

- The customer's creation growth is less so it needs to increase year by year.
- It has more products quantity available under lower criteria of discount rates so increase in discount leads to raise the sales and minimize the inventory availability.
- Too many orders cancelled by the customers which leads to reduce business growth. Hence it is better to focus on operational efficiency and customer engagement.
- DHL has one consignment in many cities except Hyderabad & Bangalore so it is required to enlarge multiple consignments to enhance the business deals.
- Inventory is sufficient but there is a possibility of shortage in the future for some products (like id 204). It should maintain by increasing the Product quantity availability after sales.
- Products like 201 sold but not shipped to the addresses like Bangalore and New Delhi so keep an eye with logistics to make deliveries timely.
- Adults on afternoon reported more cases so need to train them how to safeguard themselves from the offenders.